The empirical recurrence for OEIS A283279: recovering a conjecture that is not written down

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Abstract

OEIS A283279 records "Empirical recurrence of order 69", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 102$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283279 is "Number of nX5 0..1 arrays with no 1 equal to more than two of its horizontal, vertical and antidiagonal neighbors.". It has offset 1 and begins

$$32, 614, 10299, 191025, 3455416, 62640737, 1136604611, 20614698223, \ \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Mar 04 2017 11:53:07, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 646$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 102$ of the 646, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 102$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 204$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{69} c_i \, a(n-i),$$

c_1	13	c_{19}	4972363741	c_{37}	148814725	c_{55}	−3428535
c_2	84	c_{20}	−5560269723	c_{38}	−109486132	c_{56}	−1090000
c_3	239	c_{21}	5635671583	c_{39}	361755417	c_{57}	2596976
c_4	−1142	c_{22}	−1414388836	c_{40}	34884667	c_{58}	−1458939
c_5	−3014	c_{23}	880666737	c_{41}	81958785	c_{59}	−34975
c_6	−3254	c_{24}	−2641459213	c_{42}	71013924	c_{60}	495157
c_7	73964	c_{25}	−3348281847	c_{43}	−272526735	c_{61}	−271356
c_8	−99130	c_{26}	1641100604	c_{44}	70078037	c_{62}	22564
c_9	133412	c_{27}	5913762489	c_{45}	39455277	c_{63}	44811
c_{10}	−2454668	c_{28}	2868530705	c_{46}	−48642227	c_{64}	−24567
c_{11}	9805540	c_{29}	−4964059143	c_{47}	30119694	c_{65}	5095
c_{12}	−26310731	c_{30}	−4975045669	c_{48}	1937478	c_{66}	−152
c_{13}	99654053	c_{31}	−512002150	c_{49}	−561379	c_{67}	189
c_{14}	−381842549	c_{32}	5149077032	c_{50}	−6419094	c_{68}	−168
c_{15}	1128928590	c_{33}	1459987629	c_{51}	6116124	c_{69}	28
c_{16}	−2628690567	c_{34}	−816597124	c_{52}	−1896289		
c_{17}	4626578138	c_{35}	−1156049906	c_{53}	−3817301		
c_{18}	−5561112080	c_{36}	−566870809	c_{54}	6670620		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A283279.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.